Fiscal Year:	FY 2010	**Task Last Updated:**	FY 09/28/2009
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Project Title:	Tissue-specific acute and late molecular surveillance of particle radiation effects		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
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Space Biology Special Category:	None		
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Zip Code:	94025-3453	**Congressional District:**	18
Comments:			
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:			
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Task Description:	The space atmosphere is comprised of wide spectrum of highly energetic and charged particle radiation. Exposure to this type of radiation in the galactocosmic environment or unplanned solar particle events (SPE) may pose a potential health hazard to humans in long-term manned space explorations. A large database of information is available on the mutagenic potential of particle radiation on cultured cells, however, limited information is available for multicellular tissue responses. This is primarily due to constraints in the selection of adequate markers in the appropriate tissue systems. Transgenic rodent mutation model systems are widely accepted by regulatory agencies as an appropriate in vivo systems to access the potential carcinogenicity of a variety of test articles. In these systems, multiple copies of the inert reporter gene are integrated into every cell of the genome, making it possible to use it as a tool to measure, both quantitatively and qualitatively, mutations in every tissue of the animal. In our studies, we are using one of the transgenic mouse mutation model systems to evaluate the immediate and long-term mutagenic potential of simulated		

	space radiation beams in different tissue. Knowledge gained in this ground-based work is directed at understanding how tissue and organ respond to exposures both on the lunar surface as well as in deep space. Information gathered from this research will aid in reducing the uncertainties associated with predicting risks both in the estimation of risk to astronaut health and the design of appropriate countermeasures to ameliorate any potential negative effects in space flights.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The space atmosphere is comprised of particle radiation in a wide spectrum of highly energetic and charged (HZE). Exposures to these HZE radiations poses a serious health hazard to humans in long-term manned space explorations. We are using the plasmid-based lacZ transgenic mouse mutation model system to examine the acute and long-term tissue specific mutagenic potential after exposure to radiation with different qualities. Knowledge gained in this ground-based work is directed at understanding tissue and organ response to space radiation and will aid both in the estimation of risk to astronaut health and the design of appropriate countermeasures to ameliorate the negative effects in space flights.
Task Progress:	C57lacZ animals that are either Trp53 wild type, heterozygous or nullizygous were exposed to a range of doses of low dose-rate Solar Particle Event (SPE) simulated radiation at the NSRL at Brookhaven National Laboratory. We noted a dose-dependent increase in the # of total aberrations in bone marrow lymphocytes in the wild-type animals after 100 and 200 cGy that was significantly different from the sham-treated controls at 10 days and 4 wks post SPE irradiation. However, the level of aberrations appeared to return to normal control levels by 8 wks post irradiation, suggesting that either the damaged cells were eliminated from the early progenitor cell population in the bone marrow or that the cellular damages in this population were repaired. On the other hand, FISH results of the samples obtained from the same animals using chromosomes #2 and #8 probes revealed chromosome specific lesions that persisted even after 8 wks. Therefore, even though conventional techniques suggest that the level of chromosome aberrations in bone marrow lymphocytes at 8 weeks post irradiation was comparable to the control levels, aberrations such as translocation and insertions are not easily accessible through this technique and may be missed. Total chromosome aberrations were evaluated in bone marrow lymphocytes harvested from nullizygous animals that were exposed to SPEs. Although Tp53 is known to play a critical role in cell cycle/apoptosis and ultimately carcinogenesis, our current results appear to show that animals lacking Tp53 may not accumulate excess aberrations when compared to animals with normal functional Tp53. We evaluated the expression of neurotrophin and receptors known to be functionally significant in normal signaling pathways in hippocampal tissues collected from SPE irradiated animals from 10 days up to 8 wks after SPE exposure. Our results clearly indicate that the signature of dose-specific neurotrophin-related genes involved not only evolves with time after irradiation but is also dependent on the radiation dose. Using this approach, we can extrapolate a hierarchy of neuronal signal transduction in this tissue after low dose/low dose rate proton exposure. We also profiled the expression of genes known to be important for the onset, development and progression of Alzheimer disease in hippocampal tissues obtained from mice exposed to SPE irradiation. Surprisingly, out of the 84 genes evaluated, 5 genes in this array were consistently after animals were exposed to 50 cGy of SPE. On the other hand, additional candidate genes were involved when animals were exposed to a higher 100cGy dose of SPE. With these results, we believe that the concept of dose response must be expanded to not only include quantitative changes in the level of responses, but also qualitative changes in the profile of genetic expression. We pioneered in vivo studies exposing animals to sequential protons and iron ions. Cohorts of animals were exposed first to different doses of iron ions, followed by another dose of protons, separated by several hours between exposures. Comparison of early changes in % of micronucleated reticulocytes (MN-RET) in the peripheral blood showed no difference between animals that were irradiated with iron followed by protons and those that were irradiated with protons followed by iron ions. Earlier reports from Zhou et al (Rad Res 2006) with primary human neonatal fibroblasts show that exposing cells first to protons followed shortly (between 2.5 min to 1 hr) with iron or titanium beam resulted in more than additive effect for clonogenic survival and frequency of anchorage-independent growth. Additional evaluations of additional endpoints are in progress in our laboratory. We analyzed changes in functional genes known to be involved in transporters and metabolizing enzymes in liver samples obtained from animals that were exposed to 1 Gy 1 GeV/u iron ions or 1 GeV/u protons at 10 days after irradiation. Several enzyme transporters, including the metallothioneins, P-glycoprotein family of genes, carboxylesterases, dehydrogenases, glutathione peroxidases, and oxidoreductases were significantly upregulated ($p<0.05$) while lipoxygenases and paraoxonases were down regulated after iron ions. These results not only demonstrates that particle radiation plays a role in altering liver function in mice, but also underscores the importance of radiation quality in evaluating tissue specific endpoints.
Bibliography Type:	Description: (Last Updated: 04/26/2023)
Abstracts for Journals and Proceedings	Chang PY, Bakke J, Wang A, Bunin D. "Impact of Low dose-rate solar particle event radiation in vivo." Presented at the Heavy Ion Symposium/20th annual Space Radiation Investigator's Workshop, Koln, Germany, July 5 – 10, 2009. Proceedings of the Heavy Ion Symposium/20th annual Space Radiation Investigator's Workshop, 2009. , Jul-2009
Abstracts for Journals and Proceedings	Chang PY, Bakke J, Bunin D. "Low dose-rate Simulated Solar Particle Event Radiation in vivo." 55th Radiation Research Society Meeting, Savannah, Georgia, Oct 3 – 7, 2009. Proceedings of the 55th Radiation Research Society annual meeting, October 2009. , Oct-2009
Abstracts for Journals and Proceedings	Doppalapudi R, Bakke J, Davis Z, Menda S, Chang P. "Chromosomal Aberrations in Mice Irradiated with Protons. " 40th Environmental Mutagen Society meeting, St. Louis, Missouri, Oct 24- 28, 2009. Proceedings of the 40th Environmental Mutagen Society meeting, October 2009. , Oct-2009
Articles in Peer-reviewed Journals	Blakely EA, Chang PY. "Biology of charged particles." The Cancer Journal, 2009 Jul-Aug;15(4):271-84. PubMed PMID: 19672143 , Jul-2009